ESTR1002 Fall 2025

Lab Exercise #4

Due: 8 Oct 2025

Separate your answer for each question in a different .c file. Compress all files into a .zip and submit it to Blackboard.

Question 1

Write a C program to read a non-negative integer and count the number of '1' in all its digits..

Input: an integer in the range [0, 5000000].

Output: an integer representing the number of occurrences of the digit '1'.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
1110
3
```

Sample Run #2:

```
123
1
```

Question 2

You are given a positive integer N. Write a program that prints an N-order decreasing triangle, following these requirements:

- The triangle contains N lines and N columns.
- In the i-th line/column, there are (N-i+1) integers.
- In each row/column, the former integer is bigger than the later integer by 1.
- The first integer in the first line is N.
- The integers in each line are separated by commas (",").

Input: an integer N in [1, 15].

Output: the N-order decreasing triangle in a N-line string.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
3
3,2,1
2,1
1
```

Sample Run #2:

```
5
5,4,3,2,1
4,3,2,1
3,2,1
2,1
1
```

Question 3

Given two integers, where (i) the first integer is 0,..,6 to indicate Sunday, Monday, …, or Saturday as the first date of the month, and (ii) the second integer indicates the number of days in the month, you need to print the calendar of the month.

Input: an integer in [0, 6] and another integer in [28, 31].

Output: the calendar of the month.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
0 28
Su Mo Tu We Th Fr Sa
 1  2  3  4  5  6  7
 8  9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
```

Sample Run #2:

```
3 31
Su Mo Tu We Th Fr Sa
          1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

Question 4
Read a positive odd integer x from the user. Then, print the big star pattern. The following are 2 sample runs, where the underlined italic characters are the inputs from the user.

```
3
x..x..x
.x.x.x.
..xxx..
xxxxxxx
..xxx..
.x.x.x.
x..x..x
```

```
5
x....x....x
.x...x...x.
..x..x..x..
...x.x.x...
....xxx....
xxxxxxxxxxx
....xxx....
...x.x.x...
..x..x..x..
.x...x...x.
x....x....x
```

Question 5
Read a positive odd integer x from the user. Then, print the big A pattern. The following are 2 sample runs, where the underlined italic characters are the inputs from the user.

```
3
........*
.......*.*
......*...*
.....*.....*
....*********
...*.........*
..*...........*
.*.............*
*...............*
```

```
1
....*
...*.*
..*****
.*.....*
*.......*
```

Question 6

Complete the following C program to read a positive integer at most 100 from the user. After that, the program prints a pattern following the rules as stated below:

a) The number of rows and the number of columns equal to the user input.
b) Each row/column is classified as either a special row/column or not.
c) Suppose the leftmost column is the 1st column. The (1+2+…+i)th column in the pattern is the ith special column.
d) Suppose the topmost row is the 1st row. The odd rows are the special rows.
e) A dot is printed where both row and column are not special.
f) A hashtag is printed where either the row or the column is special.
g) An at sign is printed where both the row and the column are special.

The following are 3 sample runs, where the underlined italic characters are the inputs from the user.

8
```
@#@##@##
#.#..#..
@#@##@##
#.#..#..
@#@##@##
#.#..#..
@#@##@##
#.#..#..
```

9
```
@#@##@###
#.#..#...
@#@##@###
#.#..#...
@#@##@###
#.#..#...
@#@##@###
#.#..#...
@#@##@###
```

11
```
@#@##@###@#
#.#..#...#.
@#@##@###@#
#.#..#...#.
@#@##@###@#
#.#..#...#.
@#@##@###@#
#.#..#...#.
@#@##@###@#
#.#..#...#.
@#@##@###@#
```